HPDP Results Report

Measured results, requirement compliance and open items of the Hyperspectral Payload Data Processor

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Prepared by	Shubham Draveriya
Builds tested	m7h (commit b506d30, NX_MAX 512) and m7i_1024b (commit d2eda42, NX_MAX 1024)
Hardware	AMD ZCU106, XCZU7EV-2; two 1000BASE-SX SFP modules; 1 m duplex OM3 fiber
Specification	SPEC.md v1.0 (102 shall / should requirements)
Result	102 of 102 requirements met

This report gives the measured results of the Hyperspectral Payload Data Processor: throughput, compression, the optical link, end-to-end integrity, radiation mitigation, implementation and verification coverage, followed by the status of every requirement.

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1 Summary

Area	Key measurement	Verdict
Throughput (VER-08)	76.8 Msamples/s for 600 s, 2344 scenes, 0 dropped lines	**PASS**
End-to-end integrity (VER-07)	4 of 4 scenes bit-exact at injected BER 0, 10^{-4} and 10^{-3}	**PASS**
Optical link (VER-06)	3.375×10^{12} bits PRBS-31, 0 errors; BER < 8.9×10^{-13} at 95 %	**PASS**
Eye scan (LNK-10)	Open eye both directions: 77.5 % horizontal, 90.5 to 96.8 % vertical	**PASS**
RAW overload (SYS-09)	Raw data exceeds the link: 99.1 % of lines dropped and counted, all compressed scenes delivered	**PASS**
Compression	63 of 63 real scenes lossless; 4.4 to 6.0 bits/sample mean per dataset	**PASS**
Radiation (RAD-01 to 07)	251 configuration upsets on the board, all handled; 0 silent in 820 protected simulated trials	**PASS**
Timing (VER-09)	All 11 clocks met; WNS +0.087 ns at 250 MHz; CDC clean	**PASS**
Compressor matrix (VER-02)	864 of 864 configurations bit-exact	**PASS**
Ethernet upload (SW-05)	1 814 528 bytes over TCP identical to the JTAG copy in 4 runs; 4 of 4 scenes bit-exact	**PASS**
Independent CCSDS 123 check (CMP-02, MDL-02)	944 of 944 images byte-identical with the NTNU model, 21 board scenes among them	**PASS**

1.1 Test set-up

Item	Configuration
Board	AMD ZCU106 rev. with an x16 DDR4 SODIMM (needs the DDR override of docs/m0)
Optics	SFP0 → 1 m OM3 → SFP1, 1000BASE-SX modules, both ends on the same 156.25 MHz reference (Si570, MGTREFCLK1 of bank 226)
Clocks	clk_sys 250 MHz, clk_sensor and clk_ctrl 100 MHz, clk_link 62.5 MHz, clk_fr 31.25 MHz
Software	hpdp_run (bare metal, Arm A53, job/result format v7); host scripts run_m4.py and run_m5.py; ground station hpdp_gs.py
Loading	JTAG only; the QSPI flash was never written
Tools	Vivado / Vitis 2025.2, Python 3 with pytest

2 Throughput

Measurement	Result
VER-08: mission scene 512 × 256 × 150, D = 12, one line per ms, 2344 scenes (builds m5, m7g and m7h)	600.1 s at **76.8 Msamples/s**; 0 dropped lines, 0 underruns or overruns; 37.3 million science frames; 0 code errors; 0 sync losses; identical frame counts on all three builds
M4 capture runs (compressor output straight to DDR)	98.8 to 99.8 Msamples/s, the maximum of the sensor emulator
Compressor capacity	One sample per clock at 250 MHz (250 Msamples/s)
SYS-09: RAW mode (no compression) at the mission rate	Raw data exceeds the link as required: 99.1 % of lines dropped and counted; every compressed scene that reached the compressor was delivered
Link capacity	About 857 Mbit/s for packets; compressed data at the mission rate needs about 440 Mbit/s

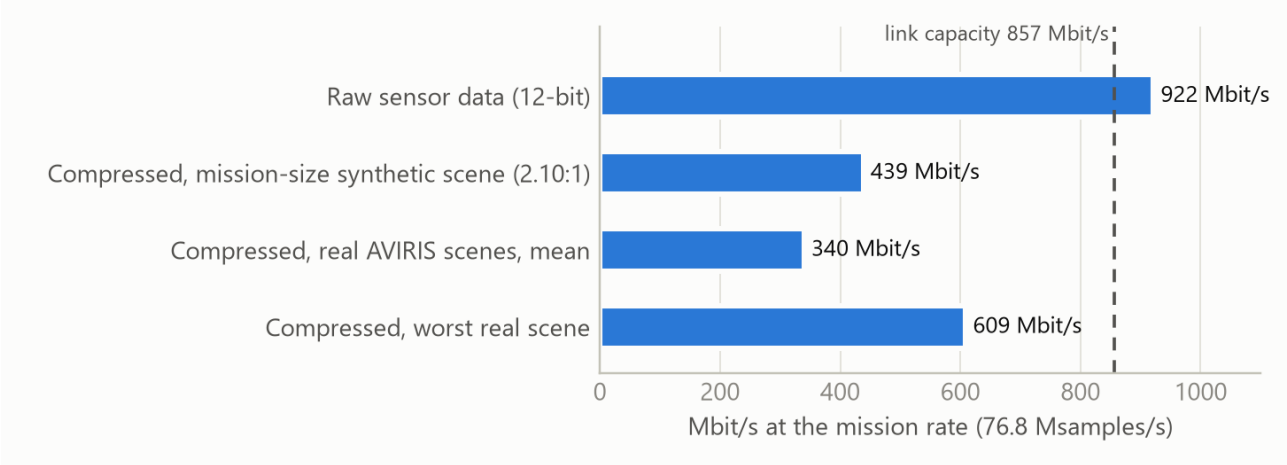


Figure 1: Data rate at the mission rate against the link capacity. Only compressed data fits; the worst real scene uses 71 % of the link.

3 Compression

3.1 Real hyperspectral data

63 public scenes went through the compressor model, which is bit-exact with the RTL over the full VER-02 matrix. Every scene was decoded and compared: 63 of 63 lossless. Five of the scenes also went through the board and were rebuilt bit-exactly at the ground over the fiber.

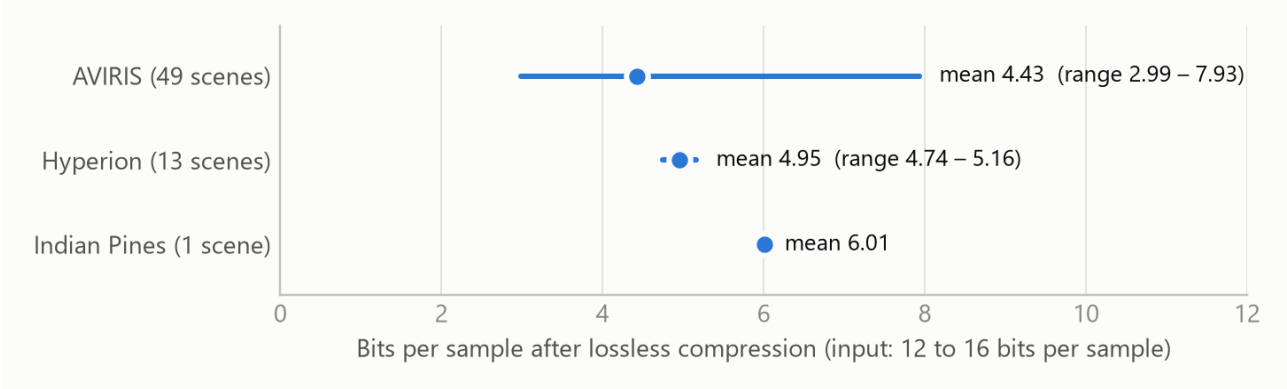


Figure 2: Bits per sample after lossless compression, default settings (P = 3, full prediction, neighbour-oriented wide local sums).

Dataset	Scenes	Bits/sample, mean (range)	Ratio	Link use, mean (worst)
AVIRIS, airborne, 512 × 512 × 224, 2008 to 2017	49	4.43 (2.99 to 7.93)	3.12:1	40 % (71 %)

Dataset	Scenes	Bits/sample, mean (range)	Ratio	Link use, mean (worst)
Hyperion, EO-1 satellite, 256 × 1024 × 242, 2017	13	4.95 (4.74 to 5.16)	2.86:1	44 % (46 %)
Indian Pines, AVIRIS, 145 × 145 × 220, 1992	1	6.01	2.33:1	54 %

- CCSDS 123 needs 1.6 to 2.0 times fewer bits than deflate on the same samples.
- AVIRIS falls into two groups: 42 scenes at 3.91 bits/sample on average, and the 7 scenes of three flights at 7.53.
- The default settings are within 1 % of the best setting tried on each dataset; without spectral prediction (P = 0) scenes need 12 to 67 % more bits.
- Scene length: 256-line scenes cost 0.3 % against whole images, 16-line scenes 2 to 4 %. The default of 256 lines stands (open item O-7 closed).

3.2 Synthetic scenes (bit-exact and decoded at the ground)

Scene	Bits/sample	Ratio
128 × 64 × 64, D = 12, spectral mixture with noise	5.92	2.03:1
512 × 256 × 150, D = 12 (mission size)	–	2.10:1
512 × 16 × 256, D = 16, P = 15 (build maximum)	10.9 to 11.1	1.45:1
Ramp pattern, D = 10	1.20	8.30:1
LFSR pattern, D = 14 (incompressible)	14.50	0.97:1

4 Optical link and channel coding

4.1 Bit error rate of the link

Test	Result
VER-06: PRBS-31 over the fiber, 2700 s	3.375 × 10^{12} bits, 0 errors: BER < 8.9 × 10^{-13} at 95 % confidence; checker locked throughout; 0 clock corrections, 0 elastic-buffer errors (common reference clock)
LNK-10: eye scans (In-System IBERT, increments 2/4, dwell BER 10^{-6})	SFP0 → SFP1: horizontal 77.5 %, vertical 96.8 %. SFP1 → SFP0: 77.5 % / 90.5 %. PRBS-31 during the scans: 1.1 × 10^{12} bits, 0 errors
Link start-up	The link locked at the first attempt in every run

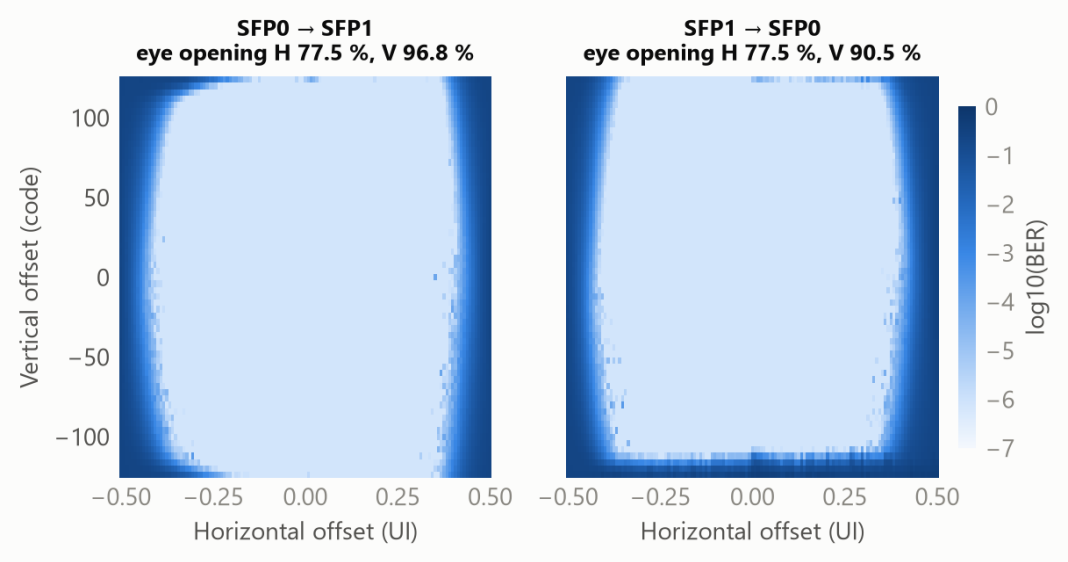


Figure 3: Statistical eye scans of both fiber directions. Light: BER at or below the 10^{-6} dwell limit; dark: high BER at the bit edges.

4.2 End-to-end integrity with injected errors (VER-07)

Injected BER	Records	Bits injected	RS symbols corrected	Uncorrectable	Scenes bit-exact
0	1772	0	0	0	4 of 4
10^{-4}	–	2187	1432	0	4 of 4
10^{-3}	–	22 200	14 508	0	4 of 4

At BER 10^{-3} the frame synchronizer stayed in LOCK throughout (60 sync-marker bit errors tolerated).

4.3 Bit error rate sweep

Injected BER	Outcome
2 × 10^{-3}	All errors corrected; scenes bit-exact
3 × 10^{-3}	One uncorrectable codeblock; it costs only its own scene (SYS-06)
4 × 10^{-3} and above	Scenes lost, as expected for RS(255,223) with I = 4 (expected limit 3 to 5 × 10^{-3})

The ground station's Reed-Solomon decoder was cross-checked against the independent *reedsolo* library on 7088 codewords. In pure Python it decodes a codeblock in 3.4 ms and a 128 × 64 × 64 scene in 2.5 s.

5 Radiation mitigation

Req.	Mechanism	Evidence on the board
RAD-01	SEM controller (sem_ultra) in correction mode, configuration scrubbing	251 configuration upsets injected: 198 corrected in 125 to 153 ms (one 154 ms scan of the configuration memory), 53 in masked or unused frames, 0 uncorrectable; scenes bit-exact afterwards
RAD-02	Built-in ECC on all 56 UltraRAMs (table, corner turn, line buffer), counters, injection	Single-bit errors corrected (1 / 4 / 4 reads); double-bit errors detected
RAD-03	Weights in block RAM with ECC; voted LUT RAM; register block in three voted copies	Counters 0 in every run, including 10 min at the mission rate
RAD-04	Control state in three copies, refreshed from the vote every cycle; extended in m7h to the corner turn, packetizer and framer	As RAD-03
RAD-05	PL pipeline watchdogs; PS watchdog per job	The PS watchdog reset the board when the software hung on purpose
RAD-06	Uncorrectable errors flag the scene's packet headers and raise an interrupt	Flagged scenes seen by the ground station
RAD-07	Fault-injection campaign	See section 5.1

5.1 Fault-injection campaign (simulation, 1300 trials on m7h)

Class	Trials	Corrected	Detected	Masked	Silent
Protected state (ECC, TMR)	820	567	238	15	**0**
Unprotected datapath registers	480	–	–	–	108 (22.5 %)

The campaign drove a design change: in build m7g, upsets in the address and count registers of the corner turn, packetizer and AOS framer were silent 93 to 100 % of the time. Build m7h triplicates them (+0.7 k LUT); afterwards all 220 simulated upsets in them were corrected. The remaining silent rate in datapath registers is accepted as residual risk.

6 Implementation

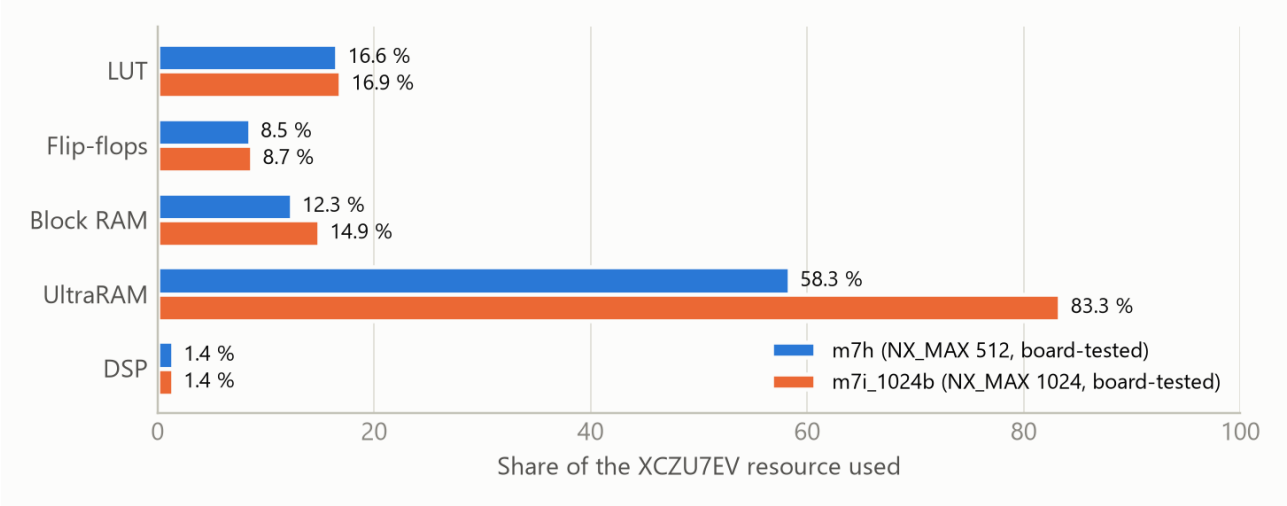


Figure 4: Device utilization of the two final builds. The 1024-pixel build is limited by UltraRAM.

Item	m7h (NX_MAX 512)	m7i_1024b (NX_MAX 1024)
LUT / FF	38.2 k (16.6 %) / 39.2 k (8.5 %)	38.9 k / 40.1 k
BRAM36 / URAM / DSP	38.5 / 56 / 24	46.5 / 80 / 24

Item	m7h (NX_MAX 512)	m7i_1024b (NX_MAX 1024)
Setup WNS / hold WHS	+0.087 ns / +0.010 ns (all 11 clocks)	+0.012 ns / +0.010 ns
Bus skew	36 constraints met, worst slack +2.94 ns	worst +2.85 ns
report_cdc	0 unsafe, 0 unknown; one documented waiver (CDC-11 inside the GTH wizard)	identical
report_methodology	5 LUTAR-1 waived (GTH wizard reset structure, as AMD's example); 2 MMCM advisories	identical

7 Verification coverage

Level	Coverage	Result
Python models	Encoder/decoder round trips over 20 configurations and D = 2 to 16, RS against reedsolo, framing, frame sync, end-to-end chain	150 tests pass
RTL regression	87 self-checking xsim tests from model vectors with random stalls: compressor configurations, corner-case images (all zero, all maximum), front end, downlink and ground blocks, link layer, system tests through the registers, ECC / TMR / watchdog / SEM paths	87 of 87 pass, 9 minutes
Configuration matrix (VER-02)	All 864 compressor configurations (local sum 4 × mode 2 × P 4 × D 3 × Ω 3 × U_{max} 3)	864 of 864 bit-exact, 57 minutes
Fault campaign	tb_fault, 1300 trials	0 silent in protected state
Board	M0 loopback; M4 15 scenes; M5 VER-06/07/08 and SYS-09; M7 runbook; real scenes, 1024 build, SD card and console, Ethernet upload	All pass, JTAG only

8 Requirements compliance

Area	Met	Not fully met
System (SYS-01 to 10)	All; SYS-04 by build m7i_1024b	–
Sensor emulator (SE-01 to 06)	All	–
Radiometric correction (RC-01 to 05)	All	–
Corner turn (CT-01 to 04)	All	–
Compressor (CMP-01 to 09)	All; CMP-02 with the NTNU model (section 9)	–
Packetizer, AOS framer	All	–
Channel coding (CC-01 to 06)	All; CC-06 with reedsolo instead of libfec	–
Error injector (EI-01 to 04)	All	–
Link (LNK-01 to 10)	All, including LNK-10 (should)	–
Ground receiver (GR-01, 03 to 05)	All (GR-02 byte-aligned)	–
Data movement (DM-01 to 03)	All	–
PS software (SW-01 to 06)	All	–
Ground station (GS-01 to 06)	All; GS-02 with reedsolo	–
Models (MDL-01 to 04)	All; MDL-02 with the NTNU model	–
Radiation (RAD-01 to 07)	All	–
Verification (VER-01 to 10)	All; VER-04 with reedsolo	–

Not done (*may* items): double-buffered tables (RC-06), near-lossless at full rate (CMP-11), hybrid entropy coder (CMP-12), hardware RS decoder (GR-06), bit-level frame sync (GR-02), C / numba decoder (GS-04). Version 1.1 item not done: near-lossless compression (CMP-10).

9 Deviations and decisions

Item	What and why
Job interface first, console later (SW-02, SW-04)	Board runs are driven by job blocks loaded over JTAG with the bitstream and the software; every run returns a result block that the host checks bit for bit. The SD card loading and UART console came after M7 and post the same job block.
NTNU model instead of WhiteDwarf / CNES (CMP-02, MDL-02, O-5)	The named decoders are not available here (ESA's repository registers only ESA member states' industry). The NTNU SmallSat Lab's open-source CCSDS 123.0-B-2 model, verified by its authors against the CCSDS test vectors, is a compressor: given our header its output is byte-identical with ours, and the coding is fully determined by image and header, so any conforming decoder decodes our streams. It found one point: full prediction with Nx = 1, now refused.
reedsolo instead of libfec (CC-06, GS-02, VER-04)	An independent RS implementation in Python with the same field and roots, available without building libfec on Windows.
Calibration table layouts (M7, SYS-04)	Layout A, one entry per 64-bit word up to 512 × 256; layout B above it, so NX_MAX 1024 fits in 80 of 96 UltraRAMs.
Test data (SPEC §11)	Indian Pines and the GICI sets of 49 AVIRIS and 13 Hyperion scenes replace Salinas and Pavia University, whose source was unreachable.
SYS-09 RAW overload	RAW mode shows that raw data exceeds the link; whole scenes are not dropped on purpose.
Scene guard	A scene cut short is padded and ended, so one lost line cannot merge scenes (found on the board in M5).
TMR on the steering state (m7h)	Added after the fault campaign.
Residual risk	Upsets in unprotected datapath registers are silent about one time in four in simulation; accepted.

10 Open items

Item	Needs	Effort
Optional: a decoder-side CCSDS 123 check	hcomp (an open-source CCSDS 123.0-B-2 decompressor) or the CNES tools	Half a day
Optional: the app refuses Nx = 1 in the console's default configuration	A check in hpdp_run and a console run on the board	An hour

The stretch goals of SPEC §15 are untouched. A return channel is now possible because the fiber is duplex (O-1 closed).

11 Conclusion

The HPDP meets its purpose: it carries hyperspectral scenes from a raw sensor stream to a bit-exact image at the ground, at the full mission rate, through a real optical link, with error correction that holds to a bit error rate of 2×10^{-3} and radiation mitigation that left no silent corruption in protected logic. All 102 requirements are met; the output is byte-identical with an independent CCSDS 123.0-B-2 implementation.

Appendix A: Reproducing the results

```
python -m pytest model                                  # Python models
vivado -mode batch -source tb/run_sim.tcl               # RTL regression (87 tests)
vivado -mode batch -source vivado/build.tcl             # bitstream + XSA
vitis -s sw/baremetal/build.py build/m7h/hpdp.xsa       # software
python sw/host/run_m5.py --work <name>                  # board run, JTAG only
python sw/ground/hpdp_gs.py decode build/runs/<name>/gr.bin --ref build/runs/<name> --show
```